



Induction of Labor

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KEY ABBREVIATIONS

American College of Obstetricians and Gynecologists	ACOG
Confidence interval	CI
Electronic fetal monitoring	EFM
Hazard ratio	HR
Hemolysis/elevated liver enzymes/low platelets	HELLP
Induction of labor	IOL
Intrauterine growth restriction	IUGR
National Institute of Child Health and Human Development	NICHD
Neonatal intensive care unit	NICU
Odds ratio	OR
Premature rupture of membranes	PROM
Prostaglandin	PG
Prostaglandin E ₁ (misoprostol)	PGE1
Prostaglandin E ₂ (dinoprostone)	PGE2
Randomized clinical trial	RCT
Relative risk	RR
Society for Maternal-Fetal Medicine	SMFM

INDUCTION OF LABOR: DEFINITION/EPIDEMIOLOGY

Induction of labor occurs by stimulating uterine contractions before the spontaneous onset of labor.¹ Induction of labor is one of the most

commonly performed obstetric procedures in the United States, with more than 23% of singleton births having undergone induction in 2012.^{1,2} Fig. 14.1, from the National Center for Health Statistics, illustrates induction of labor rates from 1990 to 2012, with current rates continuing to rise.

INDICATIONS/CONTRAINDICATIONS

The indications for induction of labor are not absolute, and induction should be undertaken only when the benefits of delivery to either mother or fetus outweigh the risks of pregnancy continuation.¹ Maternal and fetal conditions along with gestational age should be taken into account when an induction is being considered.¹ While cervical examination status should also be evaluated when considering an induction of labor, an unfavorable cervical examination should never be a reason to forgo an induction that is indicated. There are multiple accepted maternal, fetal, and obstetrical indications for labor induction (Box 14.1).^{1,3}

Box 14.2 summarizes the evaluation that should be performed before induction of labor. Indications and contraindications for induction should be reviewed, along with the alternatives, and risks and benefits of induction should be discussed with the patient. It is critical to confirm gestational age. Fetal weight should be estimated, and fetal presentation should be confirmed. In addition, a cervical examination should be performed and documented, and labor induction should be performed at a location where personnel who are familiar with the process and its potential complications are available. Uterine activity and electronic fetal monitoring (EFM) are recommended for any gravida receiving uterotonic medications. Evaluation of fetal lung

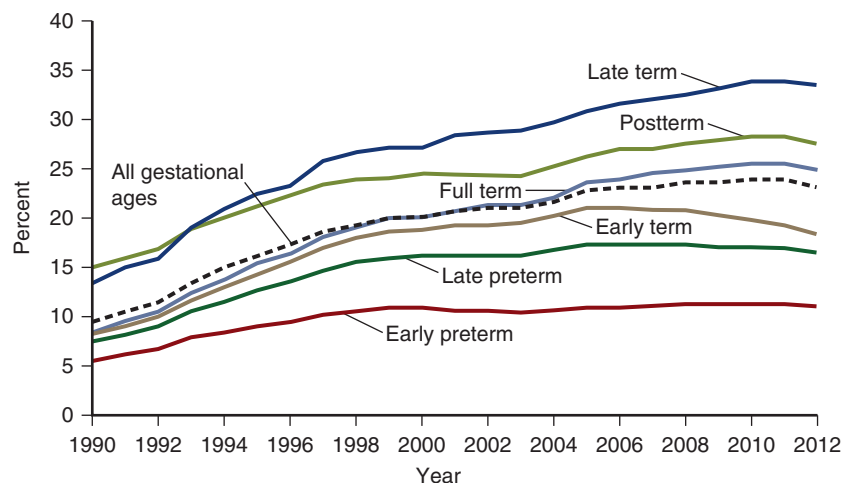


Fig. 14.1 Induction of Labor by Gestational Age: United States, 1990–2012. **NOTES:** Singletons only. Early preterm is less than 34 weeks' gestation; late preterm is 34 to 36 weeks; early term is 37 to 38 weeks; full term is 39 to 40 weeks; late term is 41 weeks; postterm is 42 weeks or more. (Data from CDC/NCHS. National Vital Statistics System. http://www.cdc.gov/nchs/data/databriefs/db155_table.pdf.)

BOX 14.1 Accepted Indications for Labor Induction

Maternal Medical Conditions

- Hypertensive disorders (gestational hypertension, preeclampsia with or without severe features)
- Chronic hypertension
- Diabetes (pregestational or gestational)
- Renal disease
- Cholestasis
- Hypercoagulable disorder (history of venous thromboembolism, acquired or inherited thrombophilia, sickle cell disease)
- Systemic lupus erythematosus

Fetal Conditions

- Fetal growth restriction
- Alloimmunization
- Oligohydramnios
- Nonreassuring fetal testing
- Multiple gestation

Obstetrical

- Postterm pregnancy
- Premature rupture of membranes
- Chorioamnionitis
- Abruptio placentae
- 39-week risk reducing
- Elective

BOX 14.2 Evaluation Before Induction of Labor

Maternal

- Confirm indication for induction.
- Review contraindications to labor and/or vaginal delivery.
- Assess the cervical condition (assign Bishop score).
- Review risks, benefits, and alternatives of induction of labor with the patient.

Fetal/Neonatal

- Confirm gestational age.
- Estimate fetal weight, either by clinical or ultrasound examination.
- Determine fetal presentation and lie.
- Confirm fetal well-being.

therefore fetal lung maturity alone should not be a reason to proceed with delivery if it could otherwise be safely delayed.

Hypertensive Disorders

In women with mild gestational hypertension or preeclampsia without severe features, investigators in the Netherlands performed a large multicenter trial in which 756 patients with singleton gestations at 36 to 41^{6/7} weeks' gestation were randomized to receive induction ($n = 377$) or expectant monitoring ($n = 379$). The primary outcome was a composite of adverse maternal outcomes and included eclampsia, HELLP (hemolysis/elevated liver enzymes/low platelets) syndrome, pulmonary edema, thromboembolic disease, placental abruption, progression to severe disease, and postpartum hemorrhage. Induction of labor was associated with a significant reduction in adverse maternal outcomes compared with expectant management, 31% versus 44% (RR 0.71, 95% CI 0.59 to 0.86, $P = .0001$). There was no difference in the frequency of neonatal complications or cesarean delivery. On the basis of these findings, **The American College of Obstetricians and Gynecologists (ACOG) Hypertension Task Force recommended, to improve maternal outcomes, delivery at or beyond 37^{0/7} weeks for mild gestational hypertension or**

maturity status is not recommended.^{3–5} The 2011 workshop titled “Timing of Indicated Late-Preterm and Early-Term Birth”⁴—sponsored by the Eunice Kennedy Shriver National Institute of Child Health and Human Development (NICHD) and Society for Maternal-Fetal Medicine—concluded that amniocentesis and assessment of fetal lung maturity should not alter delivery management if significant maternal or fetal risk warrants delivery. Moreover, documented fetal lung maturity does not confer lower nonrespiratory morbidity,^{6,7} and

preeclampsia without severe features rather than expectant management.⁸ This is discussed further in Chapter 38.

Prolonged Pregnancy

In a joint committee opinion,⁹ the ACOG and Society for Maternal-Fetal Medicine (SMFM) discouraged the use of the label *term pregnancy* and instead replaced it with the labels *early term*, *full term*, *late term*, and *postterm*. **Postterm pregnancy (≥ 42 weeks' gestation) is associated with increased risks for the fetus and mother** (see Chapter 29). According to a large epidemiologic study by Hilder and colleagues,¹⁰ the perinatal mortality rate—which combines stillbirths (fetal death after 20 weeks' gestation) and early neonatal deaths (death of a liveborn infant within the first 28 days of life)—at greater than 42 weeks' gestation is approximately twice that at term (4 to 7 deaths vs. 2 to 3 deaths per 1000 deliveries) and increases more than sixfold at 43 weeks' gestation and beyond.

Two large prospective population-based cohort evaluations of singleton pregnancies, with dating based on ultrasound, have been performed.^{11,12} Nakling and Backe¹¹ found the incidence of postterm pregnancies (≥ 42 weeks) to be 7.6%, with 0.3% of pregnancies progressing to 301 days (43 weeks' gestation) if inductions were not permitted before 43 weeks. This investigation found a significantly increased rate of perinatal mortality after 41 weeks' gestation. Heimstad and colleagues¹² also found an increased rate of intrauterine fetal demise at 42 weeks' gestation (1.55%) compared with that at 39 weeks (0.43%) and 38 weeks (0.25%), $P = .03$ for statistical trend. These studies have supported the older epidemiologic data that reported an increase in perinatal morbidity beyond 42 weeks.^{13,14} For these reasons, common practice has been to recommend delivery by 42^{0/7} weeks.

Postterm pregnancy is also associated with increased maternal risks, including an increase in labor dystocia,^{12,15} risk of cesarean,^{12,15–18} risk of hemorrhage,^{12,17} and an increase in severe perineal injury.^{16,17} Factors that may contribute to the increased rate of perinatal deaths, intolerance of labor, and risk of cesarean are uteroplacental insufficiency, meconium aspiration, fetal growth disturbances (both intrauterine growth restriction [IUGR] and macrosomia), and intrauterine infection.^{13,15,19,20}

Overall, comparisons of induction of labor versus expectant management in observational studies have found no difference in, or a decreased risk of, cesarean delivery among women who are induced.^{21–25} This is also true among women with an unfavorable cervix.²⁵ Hannah and colleagues²⁶ conducted a large multicenter trial comparing induction of labor at 41 weeks versus antenatal surveillance two to three times a week in uncomplicated singleton gestations dilated to 3 cm or less. Among 3407 women, 360 of 1701 (21.2%) underwent cesarean in the induction group compared with 418 of 1706 (24.5%) in the monitoring group ($P = .03$) with no difference in perinatal mortality and neonatal morbidity. A meta-analysis revealed that women at less than 42^{0/7} weeks' gestation who underwent induction of labor had a lower cesarean delivery rate than those who received expectant management.²⁷ In this meta-analysis, 11 randomized controlled trials (RCTs) and 25 observational studies were included and expectant management of pregnancy was associated with higher odds of cesarean delivery than was induction of labor (OR, 1.22; 95% CI, 1.07 to 1.39). Also, in a 2012 Cochrane meta-analysis²⁸ of 22 RCTs including 9383 women that compared induction of labor in term/postterm pregnancies versus expectant management, induction was associated with decreased perinatal death (RR, 0.31; 95% CI, 0.12 to 0.88), cesarean delivery (RR, 0.89; 95% CI, 0.81 to 0.97), and meconium aspiration syndrome (RR, 0.50; 95% CI, 0.34 to 0.73). On

the basis of all this information, **labor induction between 41^{0/7} weeks and 42^{0/7} weeks should be considered for women who remain undelivered by this gestational age. Induction after 42^{0/7} weeks and no later than 42^{6/7} weeks is recommended to reduce the risk of perinatal morbidity and mortality.**

Non—Medically Indicated (Elective) Induction

Non—medically indicated induction of labor refers to the initiation of labor in an individual with a term pregnancy who is free of medical or obstetric indications. Although previously not recommended, based on newer data, non—medically indicated induction of labor at or after 39 weeks is appropriate,²⁹ acceptable, and may even be routinely offered/recommended to low-risk nulliparous patients to specifically reduce maternal and perinatal risks. Conversely, non—medically indicated induction of labor before 39 weeks' gestation remains contraindicated.

The results of studies demonstrating an increased frequency of respiratory morbidity of infants induced before 39 weeks' gestation supports avoiding non—medically indicated inductions before that gestational age.^{7,30,31} A large study by Tita and colleagues of 19 centers in the Maternal-Fetal Medicine Units Network evaluated patients undergoing elective cesarean delivery without any clear medical indication at less than 39 weeks versus at or after 39 weeks. The primary outcome was a composite adverse neonatal outcome, including death, treated hypoglycemia, respiratory complications, sepsis, and admission to a neonatal intensive care unit (NICU). Birth at 37 or 38 weeks was associated with an increased risk of the primary adverse neonatal outcome (Fig. 14.2).³⁰ A recent secondary analysis of elective early-term births in a large observational study demonstrated that there were worse neonatal outcomes even with confirmation of pulmonary maturity.⁷

“Impending” macrosomia, a favorable cervix, an increased risk for preeclampsia (such as having a prior history of preeclampsia), or concerns about intrauterine growth restriction (e.g., a fetus with an estimated weight at the 19th percentile) are *not* accepted medical indications for induction before 39 weeks. Additionally, preterm or early-term induction is not medically indicated for maternal anxiety or discomfort related to normal pregnancy, previous pregnancy with labor abnormalities such as rapid labor or shoulder dystocia, or simply because the mother lives far from the hospital.

Previously, concerns have been raised that induction of labor increases the rate of cesarean delivery and health care costs.^{32,33} However, these conclusions were based on a faulty comparison of women who are induced with those in spontaneous labor.¹⁶ In fact, most observational studies that compare induction of labor with expectant management—which is the actual clinical alternative to labor induction—have found either no difference or a decreased risk of cesarean delivery among women who are induced.^{21,27} In addition, several RCTs have been undertaken. In a systematic review by Caughey and colleagues,²⁷ nine RCTs were evaluated in which women undergoing expectant management were compared with those electively induced. This meta-analysis revealed an increased risk of cesarean delivery in the expectant management group compared with the induction group (RR, 1.22; 95% CI, 1.07 to 1.39). In an observational study, Caughey and colleagues³⁴ also examined the risk of cesarean delivery for each week of gestational age ranging from 38 to 41 weeks. In their retrospective study, which again compared induction with expectant management, cesarean delivery was decreased in the induction groups. With regard to perinatal outcomes, it appears that fewer electively induced infants have meconium passage and therefore likely have a reduced incidence of meconium aspiration

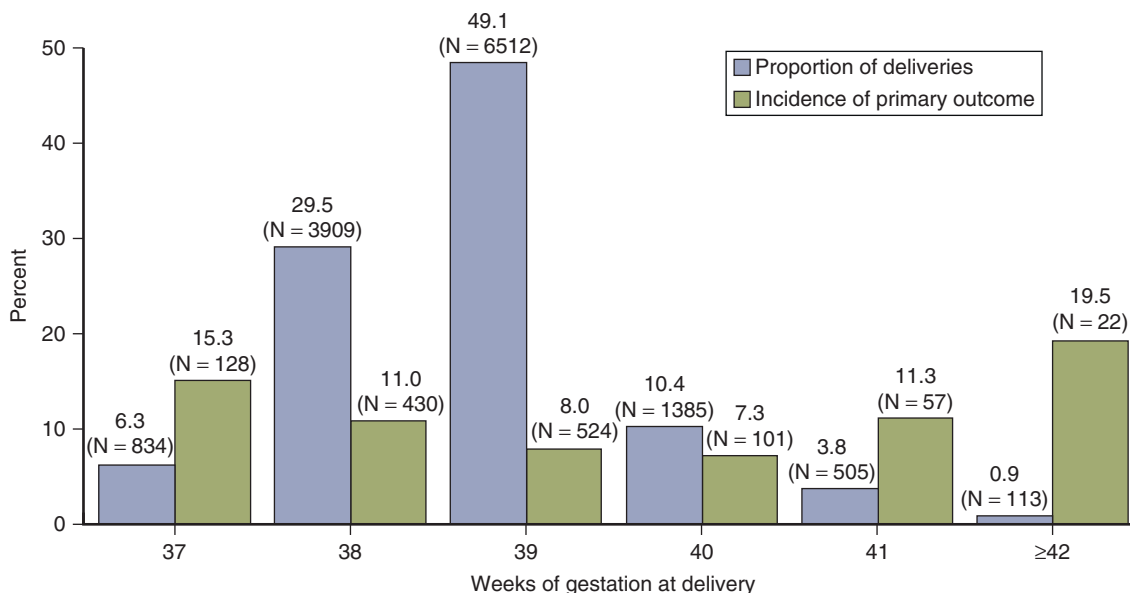


Fig. 14.2 Weeks of Gestation at Delivery. (From Tita AT, Landon MB, Spong CY et al. Timing of elective repeat cesarean delivery at term and neonatal outcomes. *N Engl J Med*. 2009;360[2]:111-120.)

syndrome.²⁷ Macrosomia may also be reduced, as noted in an ecologic study performed by Zhang and colleagues.³⁵ This study revealed that the increased induction rate between 1992 and 2003 (14% to 27%) was significantly associated with reduced mean fetal birthweight ($r = -0.54$; 95% CI, -0.71 to -0.29) and rate of macrosomia ($r = -0.55$; 95% CI, -0.74 to -0.32).

Risk-Reducing IOL in the 39th Week

The ARRIVE study, performed by Grobman and colleagues³⁶ on behalf of the NICHD Maternal-Fetal Medicine Units Network, included 6106 nulliparous women at 41 hospitals with no medical indication for delivery who were randomized to either planned induction of labor at 39^{0/7} to 39^{4/7} weeks or expectant management. The primary outcome was a composite of adverse perinatal events. The proportion of neonates with the primary perinatal outcome was lower in the induction group than in the expectant-management group (4.3% vs. 5.4%, RR, 0.80; 95% CI, 0.64 to 1.00; $P = .049$). The rate of cesarean delivery was also decreased in the induction of labor group (18.6% vs. 22.2%, RR, 0.84; 95% CI, 0.76 to 0.93) along with the rate of preeclampsia/gestational hypertension (9.1% vs. 14.1%; RR, 0.94; 95% CI, 0.56 to 0.74). There were no differences in the magnitudes of association of induction with either the primary perinatal outcome or cesarean delivery regardless of cervical status, body mass index (BMI), maternal age, or race/ethnicity. The number needed to treat to prevent one cesarean was 28, to prevent one baby requiring neonatal respiratory support was 83, and to prevent one hypertensive disorder was 20. This large randomized trial suggests that offering induction of labor to nulliparous women after 39^{0/7} weeks is acceptable and may, in fact, be associated with decreased maternal and perinatal risks. Gharvey and Macones³⁷ have argued that, on the basis of the findings of the ARRIVE trial, the term risk-reducing inductions should be considered when specifically discussing IOL with nulliparous patients in the 39th week. It is important to consider a systematic approach that ensures all eligible patients are potentially benefiting from a risk-

reducing induction and that disparities are not created or widened by differentially offering and discussing this with patients.

A secondary analysis from the ARRIVE trial found that approximately 30% ($n = 728$) of those in the expectant management arm ($n = 2502$) ultimately required a medically indicated induction.³⁸ The frequency of medically indicated induction of labor and the risk of cesarean increased as gestational age increased from 39 weeks to 41 to 42 weeks. This information may be helpful when counseling patients around risk-reducing induction in the 39th week.

PREDICTION OF LABOR INDUCTION SUCCESS/FAILURE

The cervical examination status, often referred to as the Bishop score,³⁹ was developed by Bishop in 1964 as a prelabor scoring system to assess the likelihood of going into spontaneous labor; it was not originally intended to be used to evaluate the likelihood of success of induction. The Bishop score is the system most commonly used in clinical practice in the United States to evaluate the cervix before induction. It tabulates a score based on the station of the presenting part and four characteristics of the cervix: (1) dilation, (2) effacement, (3) consistency, and (4) position (Table 14.1).^{39,40} The modified Bishop score, which gives double the amount of points for cervical dilation, was developed by Lange and colleagues⁴⁰ in 1982 and incorporates only three components of the original Bishop score (station, cervical dilation, and cervical length) (Table 14.2). The modified Bishop score has since been used in numerous trials on induction of labor.^{18,25,41-45}

Studies have evaluated cutoffs to determine favorability of the cervix before induction of labor. In general, a Bishop score of 6 or below is considered unfavorable, and cervical ripening should be considered before starting an induction of labor since these women are at higher risk for a cesarean.⁴⁵⁻⁴⁷ A Bishop score of 8 or above means that the probability of a vaginal delivery after an induction is equivalent to spontaneous labor at the same gestational age and that the

TABLE 14.1 Traditional Bishop Score System

Parameter	SCORE			
	0	1	2	3
Dilation (cm)	Closed	1–2	3–4	≥6
Effacement (%)	0–30	40–50	60–70	≥80
Length (cm)	≥3	2–3	1–2	<1
Station	–3	–2	–1, 0	+1, +2
Consistency	Firm	Medium	Soft	—
Position	Posterior	Midposition	Anterior	—

TABLE 14.2 Modified Bishop Score System

Parameter	SCORE			
	0	1	2	3
Station (cm)	–3	–2	–1, 0	+1, +2
Effacement (%)	0–30	40–50	60–70	≥80
Length (cm)	≥3	2–3	1–2	<1
SCORE	0	2	4	6
Dilation (cm)	0	1–2	3–4	>4

value of cervical ripening is less clear.¹ It should be noted, however, that although an unfavorable cervix implies an increased chance of a cesarean after labor induction, it does not signify that the risk is raised specifically because of the labor induction. As noted earlier, an unfavorable cervix (e.g., at term) has been demonstrated in randomized trials to serve as a more general marker of the risk of cesarean delivery regardless of whether labor induction or expectant management is undertaken.³⁶

Numerous observational studies have evaluated risk factors for cesarean among women undergoing an induction of labor. The most common demographic and clinical risk factors identified are older maternal age, higher BMI, greater gestational age, shorter height, nulliparity, higher neonatal birthweight, and unfavorable cervical examination.^{45,48–50} These are, however, risk factors for cesarean among women who present in spontaneous labor as well, thereby limiting the ability to truly predict risk of cesarean specifically for induced women.⁴⁵

A number of studies have evaluated the utility of combining known demographic and clinical factors in an effort to predict the success of labor induction,^{43,50–53} with many investigators creating models and nomograms to predict induction success. These models have varying degrees of accuracy (area under the curve ranging from 0.71 to 0.81), with most identifying a favorable cervical examination as a key driver. Few have been externally validated.

In one study, Levine and colleagues⁴³ evaluated risk factors for cesarean after induction, specifically limiting their analysis to term singleton pregnancies with an unfavorable cervical examination (Bishop score ≤6 and cervical dilation ≤2 cm). They also excluded

anyone with ruptured membranes. They found that the following five factors were the strongest independent predictors of cesarean delivery: shorter maternal height, increasing BMI, nulliparity, gestational age older than 40 weeks, and modified Bishop score. They derived a predictive model and nomogram (Fig. 14.3) that went through both internal and external validation. This model was converted into a user-friendly online calculator, which can be found online at <http://www.uphs.upenn.edu/labor-induction-calculator/>. It should be noted that this calculator provides information as to the risk of cesarean with induction but does not convey who will have a lower chance of cesarean with expectant management.

Additional work by this group found the use of the calculator, once an individual arrived on labor and delivery for their IOL, was associated with a 6% absolute risk reduction in maternal morbidity (17.9% to 11.6%, $P < 0.001$), as well as an 8% absolute risk reduction in cesarean delivery (28.9% to 20.3%, $P < 0.001$).⁵⁴ They hypothesized that knowledge of the calculator score assisted clinicians in decision making, particularly for those with very low or very high cesarean risk scores (e.g., performing the cesarean sooner for those with a high risk score and thereby decreasing prolonged labor and its associated morbidity).

Other attempts to use biomarker and sonographic assessments to predict the success of induction have had limited success. Elevated fetal fibronectin (fFN) concentration in cervicovaginal secretions has been thought to represent a disruption or inflammation of the chorionic-decidual interface and could plausibly be associated with successful induction. Nevertheless, studies that have evaluated fFN have demonstrated its limited ability to add substantively to the capacity of other clinical markers to predict induction success.^{44,55,56} For example, although Crane and colleagues⁴⁴ found that fFN was associated with successful induction, it was not determined to be superior to Bishop score. Cervical length by ultrasound assessment at the time of induction has also had mixed results,^{44,57–59} with no studies demonstrating definitive improvement in prediction accuracy over cervical examination and Bishop score alone. The lack of clinical utility over Bishop score alone was confirmed in a recent large meta-analysis of 31 prospective studies that included 5029 women.⁵⁹

CERVICAL RIPENING AND INDUCTION METHODS

Cervical ripening, or “softening of the cervix,” is a required process before both spontaneous labor and iatrogenic initiation of labor. The cervix is made up of fibrous connective tissue and an extracellular matrix that consists of collagen (mostly types I and III), elastin, proteoglycans, fibroblasts, and vasculature.⁶⁰ The exact process that leads to cervical ripening is not fully understood. It is a complex process that results in physical softening and distensibility of the cervix and ultimately leads to partial cervical effacement and dilation.^{60,61} Remodeling of the cervix involves enzymatic dissolution of collagen fibrils, an increase in water content, and chemical changes. These changes are induced by hormones (estrogen, progesterone, relaxin), cytokines, prostaglandin (PG), and nitric oxide synthase enzymes.⁶⁰

Cervical ripening methods fall into two main categories: *pharmacologic* and *mechanical* (Box 14.3). A cervical examination is essential before determining which, if any, method to use for cervical ripening.

Pharmacologic Methods of Cervical Ripening

Prostaglandins

Administration of PGs results in the dissolution of collagen bundles and an increase in the submucosal water content of the cervix. These changes in cervical connective tissue at term are similar to those

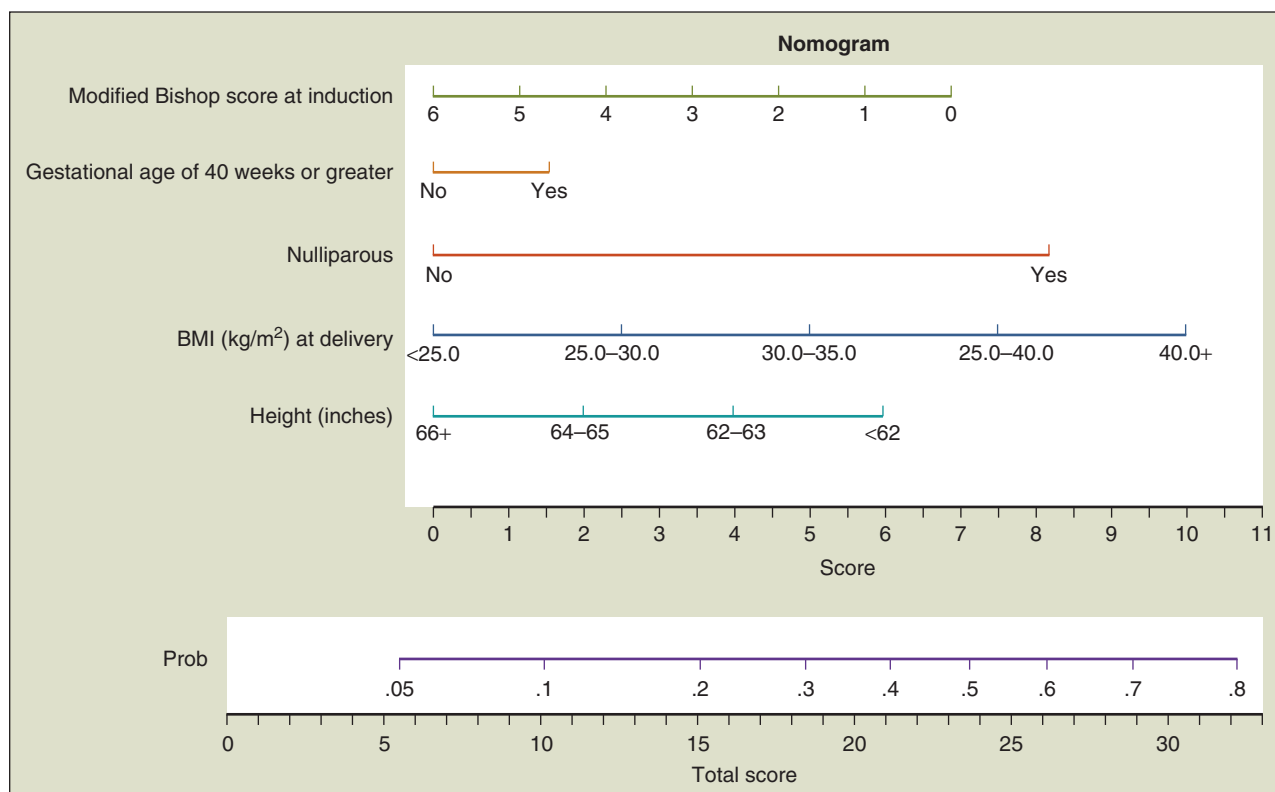


Fig. 14.3 Nomogram. BMI, Body mass index. (From Levine LD, Downes KL, Parry S et al. A validated calculator to estimate risk of cesarean after an induction of labor with an unfavorable cervix. *Am J Obstet Gynecol.* 2018;218[2]:254.e1-254.e7.)

BOX 14.3 Methods of Cervical Ripening

Pharmacologic

- Prostaglandin E₂ (dinoprostone)
- Synthetic prostaglandin E₁ (misoprostol)

Mechanical

- Cervical Foley balloon catheter
- Stripping membranes
- Hygroscopic dilators
- Extraamniotic saline infusion
- Amniotomy

observed in early labor. PGs are endogenous compounds found in the myometrium, decidua, and fetal membranes during pregnancy. The chemical precursor is arachidonic acid. PG formulations have been used since they were first synthesized in the laboratory in 1968, and PG analogues were originally given intravenously and orally. Later, local administration of PGs in the vagina or the endocervix became the route of choice because of fewer side effects. Side effects of all PG formulations and routes may include fever and chills, and their impact on the smooth muscle of the gastrointestinal track may lead to an increase in nausea, vomiting, and diarrhea.⁶²

The efficacy of locally applied PGs (vaginal or intracervical) for cervical ripening and labor induction has been demonstrated in systematic reviews that included more than 10,000 women.^{63,64}

The various administration vehicles for PG (tablet, gel, and timed-release pessary) appear to be equally efficacious in achieving a vaginal delivery, although differences in time to delivery have been noted. The optimal route, frequency, and dose of PGs of all types and formulations for cervical ripening and labor induction have not been determined. Also, PG formulations of any kind should be avoided as third-trimester cervical ripening agents in women with a prior uterine scar, such as a prior cesarean delivery or myomectomy, because their use has been associated with an increased risk of uterine rupture.^{1,65} Uterine activity and fetal heart rate monitoring are indicated for 0.5 to 2 hours after administration of PGs for cervical ripening and should be maintained as long as regular uterine activity is present.¹ Regular uterine activity is often not clearly defined. However, if the patient is having painful contractions more than three times in 10 minutes, caution should be used when additional doses of PG are being considered.

Prostaglandin E₂

Multiple trials have evaluated the effectiveness of prostaglandin E₂.^{63,64,66,67} An analysis from the 1980s summarized 59 clinical trials comprising 3313 women in which either intracervical or intravaginal prostaglandin E₂ (PGE₂, dinoprostone) was used for cervical ripening before induction of labor⁶⁷ and concluded that local administration of PGE₂ is effective in enhancing cervical effacement and dilation, shortening the induction-to-delivery interval, reducing oxytocin use, and lowering the chance of cesarean delivery for failure to progress when compared with oxytocin alone. These findings were confirmed in a Cochrane meta-analysis in 2014 of 44 trials performed worldwide using various PG compounds and dosing regimens. They found that

all forms of prostaglandin E₂, regardless of dosing and route of administration, appear to be as effective as each other in decreasing cesarean delivery rates, and all were found to have a benefit over oxytocin alone. Because no differences in clinical outcomes are apparent when comparing intravaginal or intracervical PGE₂ preparations, vaginal administration is the preferred route given its greater ease of administration, fewer gastrointestinal side effects, and patient satisfaction.⁶⁴

Currently two PGE₂ preparations have been approved by the US Food and Drug Administration (FDA) for cervical ripening—one has a cervical administration (*Prepidil*) and the other has a vaginal administration (*Cervidil*).¹ *Prepidil* contains 0.5 mg dinoprostone per 3-g syringe (2.5-mL gel) and is administered intracervically. It is stored in the refrigerator but administered at room temperature. The dose can be repeated in 6 to 12 hours if cervical change is inadequate and uterine activity is minimal following the first dose. Oxytocin should not be initiated until at least 6 hours after the last dose because of the potential for uterine tachysystole with concurrent oxytocin and PG administration. *Cervidil* is a vaginal insert that contains 10 mg of a slow-release suppository that releases 0.3 mg of dinoprostone per hour and may be left in place for up to 12 hours and then repeated. This medication is stored in the refrigerator and must be placed shortly after refrigerator removal as it becomes unstable at room temperature. A benefit of this insert is that it may be removed with the onset of active labor, rupture of the membranes, or development of uterine tachysystole; however, it is relatively expensive compared to prostaglandin E₁. Per the manufacturer's instructions, oxytocin may be initiated 30 to 60 minutes after removal of the insert. Lubricants should be avoided when placing as they can prevent the release of the dinoprostone from the vaginal insert.

Prostaglandin E₁

Misoprostol is a synthetic PGE₁ analog available as 100- and 200-μg tablets. The current FDA-approved use of misoprostol is for the treatment and prevention of peptic ulcer disease related to the chronic use of nonsteroidal antiinflammatory drugs (NSAIDs). Additionally, the FDA has approved the relabeling of misoprostol, making its use appropriate for cervical ripening and for the induction of labor during pregnancy.⁶⁸ **Administration of misoprostol for preinduction cervical ripening is considered a safe and effective use by ACOG.** Misoprostol is inexpensive and is stable at room temperature; it can be administered orally, buccally, or placed vaginally with few systemic side effects. Although not scored, the tablets can be divided to provide 25- or 50-μg doses.¹

A large body of published reports, including two large network meta-analyses, support the safety and efficacy of misoprostol when used for cervical ripening and labor induction^{56,69} in the inpatient setting. Multiple studies suggest that misoprostol tablets placed vaginally are either superior in efficacy or equivalent to intracervical PGE₂ gel in reducing cesarean delivery risk. More recently, a meta-analysis of 70 trials⁶⁴ revealed the following points regarding the use of misoprostol compared with other methods of cervical ripening and labor induction. Misoprostol improved cervical ripening compared with placebo and was associated with a reduced failure to achieve vaginal delivery within 24 hours (RR 0.36; 95% CI, 0.19 to 0.68). Additionally, compared with other vaginal PGs for labor induction, vaginal misoprostol was more effective in achieving vaginal delivery within 24 hours (RR, 0.80; 95% CI, 0.73 to 0.87). Compared with vaginal or intracervical PGE₂, augmentation of oxytocin was also less common with misoprostol (RR, 0.65; 95% CI, 0.57 to 0.73). However, uterine tachysystole with fetal heart rate changes (RR, 2.04; 95% CI,

1.49 to 2.80) and meconium-stained amniotic fluid (RR, 1.42; 95% CI, 1.11 to 1.81) were more common with misoprostol than other prostaglandins. Most studies suggested that restricting the dose of misoprostol to 25 μg every 4 hours (compared with 50 μg) significantly reduced the risk of uterine tachysystole with and without changes in fetal heart rate compared with higher doses. Most important, regardless of misoprostol dose, no significant differences were reported in immediate neonatal outcomes.

Although ACOG recommends 25 μg every 3 to 6 hours with vaginally applied misoprostol, the optimal dose and timing interval is not known.¹ If necessary, oxytocin may be initiated 4 hours after the final misoprostol dose.

Oral misoprostol for cervical ripening has also been studied. This route of administration has promise for offering more patient satisfaction and convenience. Most trials compared oral misoprostol dosages of 20 to 50 μg with similar vaginal misoprostol dosing regimens, such as 25 to 50 μg. This oral dosing regimen appears to be no more effective than vaginal administration for achieving vaginal delivery, but it may be associated with less uterine tachysystole. A positive dose-response relationship exists between the dosage of oral misoprostol and the frequency of tachysystole.⁷⁰ Some investigators have described titrating oral misoprostol to its desired effect.^{71–74} In one study misoprostol was dissolved to make an oral solution⁷¹ (e.g., a 200 μg tablet in 200 mL tap or sterile water, with 20 mL, the equivalent of 20 μg, given initially). The providers increased the dose by 20 μg after two or three doses of 20 μg until adequate contractions were achieved (3 per 10 minutes, lasting 30 seconds). Because oral dosing has a short (2-hour) duration of action, administration was repeated at 2-hour intervals. This method was as effective as vaginal dinoprostone with similar rates of hyperstimulation with fetal heart rate changes.

In a Cochrane review of 76 RCTs that included 14,412 women, oral misoprostol appeared to be at least as effective as other currently used methods of induction.⁷⁰ Within this Cochrane review, 12 trials (3859 women) compared oral misoprostol with vaginal dinoprostone and demonstrated a lower cesarean rate for women who used oral misoprostol (RR, 0.88; 95% CI, 0.78 to 0.99). The authors of this Cochrane review recommend that if clinicians choose to use oral misoprostol, a lower dose (e.g., 20 to 25 μg) in solution is preferred for safety considerations. When comparing oral misoprostol to Foley catheter alone for induction, a large noninferiority RCT demonstrated similar cesarean delivery risk (RR 0.84, 95% CI 0.69–1.02) and morbidity risk (aRR 1.06, 90% CI 0.86–1.31) between the two groups with a small increase in gastrointestinal side effects with oral misoprostol.⁷⁵

It is important to note that different routes of administration of misoprostol have different times to peak concentration, different peak levels, and different changes in their level over time.^{76–78} For example, oral misoprostol peaks in 12 to 28 minutes, with levels going to baseline in approximately 2 hours. Vaginal misoprostol peaks in 65 to 72 minutes, and levels continue to remain high for up to 4 hours. For this reason, it is important to be aware that the frequency of dosing is different depending on the route by which the misoprostol is administered. Most studies demonstrating the efficacy and safety of oral misoprostol use the 20- to 25-μg oral tablet or oral solution every 2 hours. Other routes of administration for misoprostol that have been used include buccal and sublingual. On the basis of only three small trials included in the Cochrane meta-analysis,⁷⁹ sublingual misoprostol appears to be at least as effective as the same dose administered orally, although data are inadequate to comment on the comparative frequency of its complications and side effects. The studies using buccal or sublingual misoprostol are more heterogeneous in their dosing regimens and are too small to determine safety. Therefore more

data are needed to clarify both the safety and efficacy of buccal and sublingual misoprostol. The current ACOG recommendation is that buccal and sublingual misoprostol not be used for cervical ripening.

Mechanical Methods of Cervical Ripening

Mechanical devices, which are placed in the lower uterine segment/cervix, likely work at least in part by causing the release of prostaglandin F_{2-α} from the decidua and adjacent membranes or from release of PGE₂ from the cervix. Additionally, dilators (e.g., laminaria, osmotic dilators) and balloon catheters physically cause gradual cervical dilation through osmotic or physical expansion. Some of the advantages of the mechanical dilation techniques include low cost, low risk of tachysystole, and few systemic side effects. The disadvantages of mechanical methods include the potential for disruption of a low-lying placenta and maternal discomfort on placement and manipulation of the cervix.

Intracervical Balloon Catheter

The intracervical Foley balloon catheter is a popular method of cervical ripening that has been safely used for decades.^{1,66,80–87} A 16- to 26-French catheter is most commonly used with either a 30-mL or 80-mL balloon. Multiple large studies have demonstrated the safety and efficacy of the intracervical Foley catheter as a cervical ripening method.^{66,80–84,88} Compared with PGs, there is a significantly decreased risk of tachysystole and fetal heart rate changes with cervical Foley use without a difference in cesarean delivery rates.^{66,83,84,86,88}

Balloon Catheter Inflation Amounts. A 30-mL Foley balloon catheter is most often used for inflation, although an 80-mL balloon is also available. Multiple studies have evaluated the impact that different inflation volumes have on outcomes, including cesarean and time to delivery.^{89–93} Although many trials suggested a shorter time to delivery, none found a substantial benefit in decreasing cesarean delivery rates. A meta-analysis⁹³ of seven randomized trials, which included 1432 singleton gestations, found a 2-hour shorter time to delivery with higher inflation amounts (60 to 80 mL vs. 30 mL) with no difference in cesarean sections (16% vs. 18%; RR, 0.84; 95% CI, 0.6 to 1.17). This meta-analysis and the individual studies did not have sufficient sample sizes to detect clinically important differences in maternal and neonatal morbidity. Also, there has not been consistent reporting of the rupture rate of balloon catheters. Lastly, it remains unknown whether the 2-hour time savings with higher inflation amounts would be present if combination methods were to be used (see section below).

Length of Time the Foley Is Left in Place. A four-arm randomized trial in China⁹² compared both the inflation amounts and the time the balloon catheter was left in place (80-mL inflation in for 12 hours vs. 24 hours and 30-mL inflation in for 12 hours vs. 24 hours). This study had two important findings: balloon removal at 12 hours (vs. 24 hours) resulted in a higher rate of vaginal delivery within 24 hours (50.4% vs. 28.5%, $P < .001$) and a shorter time to delivery (29.4 vs. 23.4 hours, $P < .001$). These findings are perhaps influenced by the fact that other interventions, including amniotomy, could occur if the balloon catheter was removed earlier, suggesting no added benefit in keeping the catheter in longer than 12 hours. When 80-versus 30-mL inflation volumes were compared among those with the catheter removed after 12 hours, there were no clinically significant differences in outcomes. Another recent trial ($n = 177$) evaluated the benefit of removing the Foley catheter at 6 hours versus 12 hours and found a shorter time to delivery (19.2 vs. 24.3 hours, $P = 0.04$) and a higher percentage of patients who delivered

in 24 hours (67.4% vs. 47.4%, $P < 0.01$) with removal at 6 hours compared with 12 hours.⁹⁴

Single- Versus Double-Balloon Catheter

A number of studies have evaluated single- versus double-balloon catheters for cervical ripening. Although one trial⁹⁵ had a shorter time to delivery with the double-balloon catheter, the women who received it required removal of the catheter more frequently owing to discomfort. Another trial⁹⁶ demonstrated a higher vaginal delivery rate (86% vs. 74%, $P = .02$) and a lower rate of an adverse composite neonatal outcome (1.4% vs. 7.4%, $P = .02$) with the single-balloon catheter. In a more recent trial⁹⁷ comparing single- versus double-balloon catheters, there appeared to be a higher rate of vaginal delivery (60% vs. 32%, $P = .047$), as well as of meconium with the double-balloon catheter (26.5% vs. 10%, $P = .04$). However, in this trial there were significant baseline differences between the groups (as in the number of women undergoing a trial of labor after cesarean: 15% of women with the single-balloon catheter had a prior cesarean compared with 4% with the double-balloon catheter), which make the validity of the results uncertain. Two recent meta-analyses^{98,99} determined that there was no clinical benefit of the double-balloon catheter over the single-balloon catheter and showed that the double-balloon catheter was associated with lower patient satisfaction. **Therefore—without clear clinical benefit, with the potential increase in morbidity and patient discomfort, and given the increased cost of the double-balloon catheter (\$200 to \$300 per catheter)—the single-balloon catheter is preferred.**

Hygroscopic Cervical Dilators

Dilapan-S (Dilapan, Medicem, Czech Republic) is a hygroscopic cervical dilator made from a synthetic hydrogel with the goal of absorbing fluid from cervical cells, leading to cell membrane dehydration and an increase in volume, thereby achieving cervical dilation. This was FDA approved for third-trimester cervical ripening in 2015. These dilators are most often placed cervically with a sponge forceps and left in place for 12 hours. Approximately two to five rods are required to achieve a similar dilation to a Foley balloon. A population-based study that included 543 patients and 11 sites found *Dilapan-S* to be safe and effective.¹⁰⁰ A randomized noninferiority trial that included 419 patients compared *Dilapan-S* to Foley catheter among term patients with an unfavorable Bishop score. The investigators found a similar vaginal delivery rate (81.3% vs. 76.1%, $P = 0.2$) with no difference in safety outcomes.¹⁰¹ Therefore *Dilapan-S* should be considered a reasonable alternative to the single-balloon catheter for mechanical dilation.

Membrane Stripping

Stripping or sweeping of the fetal membranes refers to digital separation of the chorioamniotic membrane from the wall of the cervix and lower uterine segment by inserting the examiner's finger beyond the internal cervical os and then rotating the finger circumferentially along the lower uterine segment. This action is thought to lead to an increase in phospholipase A₂ and prostaglandin F_{2α}. The process of membrane sweeping can be used alone to initiate spontaneous labor and avoid pregnancy prolongation or as an adjunctive method for women already undergoing induction of labor.

A 2020 Cochrane review aimed to assess the efficacy and safety of membrane sweeping for induction of labor in women at or near term (≥ 36 weeks' gestation).¹⁰² This review included 44 studies reporting data from 6940 patients and their infants. Forty of the studies (6548 patients) compared membrane sweeping with no treatment. In this group, membrane sweeping resulted in a higher chance of spontaneous

labor onset (aRR 1.21; 95% CI 1.08 to 1.34), a lower likelihood of induction of labor (aRR 0.73; 95% CI 0.56 to 0.94), and no difference in cesarean delivery, spontaneous vaginal birth, maternal or neonatal death, or serious morbidity. Most studies did not report on safety outcomes of tachysystole, uterine rupture, or neonatal encephalopathy. Four studies compared membrane sweeping with prostaglandins (480 patients) and found no difference in spontaneous onset of labor or vaginal delivery, cesarean delivery, and maternal or neonatal serious morbidity or mortality. Three studies reported on patient satisfaction, and overall patients were positive with membrane sweeping while acknowledging some discomfort. Although studies on the use of membrane stripping did not reveal an increase in either maternal or neonatal infection associated with the procedure, there are limited data.¹⁰² Only two small studies reported on membrane sweeping in patients who were group B *Streptococcus* (GBS) positive, and no additional risk was found in women with sweeping. GBS colonization is not a contraindication to membrane stripping because no direct evidence of harm exists.^{103,104}

A more recent meta-analysis¹⁰⁵ included four randomized trials on membrane sweeping as an adjunct to an induction and included a total of 1377 women. When added to other methods used for a formal induction (e.g., misoprostol, oxytocin), membrane sweeping increased the rate of vaginal delivery (RR, 1.12; 95% CI, 1.05 to 1.18). It is important to note that none of the four studies included in this meta-analysis used intracervical balloon ripening; therefore the utility of adjunctive membrane sweeping in women with intracervical balloons is still unknown. In summary, **since membrane sweeping has the potential to be beneficial in avoiding induction and shortening duration of pregnancy with minimal risks, it is reasonable to offer membrane stripping after 39 weeks' gestation to patients who want to hasten the onset of spontaneous labor.**

Oxytocin for Induction

Oxytocin is a polypeptide hormone produced in the hypothalamus and secreted from the posterior lobe of the pituitary gland in a pulsatile fashion. It is identical to its synthetic analog, which is among the most potent uterotonic agents known. Synthetic oxytocin is an effective means of labor induction.¹⁰⁶ Exogenous oxytocin administration produces periodic uterine contractions first demonstrable at approximately 20 weeks' gestation, with increasing responsiveness with advancing gestational age, which is primarily due to an increase in myometrial oxytocin binding sites. Little change occurs in myometrial sensitivity to oxytocin from 34 weeks to term; however, once spontaneous labor begins, the uterine sensitivity to oxytocin increases rapidly. **This physiologic mechanism makes oxytocin less effective in inducing labor than augmenting it and even less effective as a cervical ripening agent when used without other cervical ripening agents.**

Oxytocin is most often given intravenously. It cannot be given orally because the polypeptide is degraded to small, inactive forms by gastrointestinal enzymes. The plasma half-life, estimated at 3 to 6 minutes, is short, and steady-state concentrations are reached within 30 to 40 minutes of initiation or dose change. Synthetic oxytocin is generally diluted by placing 10 units in 1000 mL of an isotonic solution, such as normal saline, yielding an oxytocin concentration of 10 mU/mL. It is given by infusion pump to allow continuous, precise control of the dose administered.

Although oxytocin is an effective means of labor induction in women with a favorable cervix, as noted earlier, it is less effective as a cervical ripening agent, as demonstrated by many RCTs that have compared oxytocin with various methods of cervical ripening.^{63,64,106}

TABLE 14.3 Labor Stimulation With Oxytocin: Examples of Low- and High-Dose Regimens

Regimen	Starting Dose (mU/min)	Incremental Increase (mU/min)	Dosage Interval (min)
Low dose	0.5–2	1–2	15–40
High dose	6	3–6 ^a	15–40

^aThis incremental increase is reduced to 3 mU/min in presence of hyperstimulation and reduced to 1 mU/min with recurrent hyperstimulation.

From American College of Obstetricians and Gynecologists. Induction of labor. ACOG Practice Bulletin No. 107. *Obstet Gynecol.* 2009;114:386-397.

In a Cochrane review¹⁰⁶ of 110 trials that included more than 11,000 women and compared oxytocin with any vaginal PG formulation for labor induction, oxytocin alone was associated with an increase in unsuccessful vaginal delivery within 24 hours (52% vs. 28%; RR, 1.85; 95% CI, 1.41 to 2.43). No difference was found in the rate of cesarean delivery between the groups. When oxytocin alone was compared with intracervical PGs or intracervical Foley for labor induction, oxytocin alone was associated with an increase in unsuccessful vaginal delivery within 24 hours (51% vs. 35%; RR, 1.49; 95% CI 1.12 to 1.99) and an increase in cesarean delivery (19% vs. 13%; RR, 1.42; 95% CI, 1.11 to 1.82). Therefore oxytocin alone should not be used as a cervical ripening agent.

Oxytocin Dosing Intervals and Protocols

The optimal regimen for oxytocin administration is debatable, likely due to individual variation in sensitivity and response to oxytocin. Protocols differ as to the initial dose, incremental time period, and steady-state dose (Table 14.3).¹ Because success rates for varying protocols are similar, a single dosing strategy cannot be recommended. A maximum oxytocin dose has not been established, but most protocols do not exceed 42 mU/min.

Low-dose protocols mimic endogenous maternal physiology and are associated with lower rates of uterine tachysystole. Low-dose oxytocin is initiated at 0.5 to 1 mU and is increased by 1 mU/min at 30- to 40-minute intervals. An alternative low-dose protocol begins at 1 to 2 mU/min and increases by 2 mU/min with shorter incremental time intervals of 15 to 30 minutes.

High-dose oxytocin regimens often start with an initial oxytocin dose of 6 mU/min¹⁰⁷ that is increased by 6 mU/min at 15- to 40-minute intervals, or start at 4 mU/min with 4 mU/min incremental increases every 15 minutes. Although often used for augmentation, they have also been studied in the context of induction. A prospective observational study¹⁰⁸ that involved nearly 5000 women at Parkland Hospital comparing low- and high-dose oxytocin regimens for labor induction and augmentation was undertaken. The results indicated that there was no difference in overall cesarean delivery rate. However, among women augmented, those given the high-dose regimen had fewer cesarean deliveries for dystocia (12% vs. 9%, $P = .04$), whereas those women *induced* and given the high-dose regimen had a higher rate of cesarean delivery for fetal indications (6% vs. 3%, $P = .05$). The high-dose regimen patients who were *augmented* had a significantly shorter mean admission to delivery time, decreased rate of chorioamnionitis, and fewer forceps deliveries. Further, in patients who

received the high-dose regimen, there was less neonatal sepsis than in subjects given the low-dose regimen *for both augmented and induced groups*. Additionally, there were higher rates of uterine hyperstimulation in those receiving the high-dose regimen in both the augmented and induced groups. However, there was no difference in the 5-minute Apgar score or umbilical artery pH below 7 in either group. Merrill and Zlatnik¹⁰⁹ conducted a randomized double-blind trial that included 1307 patients and compared high-dose oxytocin (4.5 mU/min initially increased by 4.5 mU/min every 30 minutes) with a low-dose regimen (1.5 mU/min initially, increased by 1.5 mU/min every 30 minutes) for both induction and augmentation of labor. In the group that received high-dose oxytocin, labor was significantly shorter whether the drug was used for induction (8.5 vs. 10.5 hours, $P < .001$) or for augmentation (4.4 vs. 5.1 hours, $P = .3$), although no significant difference in the rates of cesarean delivery between the two regimens (15% vs. 11.3%, $P = .17$) was observed. Although more decreases or discontinuations of oxytocin were seen in the high-dose group both for uterine tachysystole and fetal heart rate abnormalities, neonatal outcomes were observed to be similar in both groups.

A 2014 Cochrane review¹¹⁰ of nine trials demonstrated that high-dose oxytocin reduced the induction-to-delivery interval but not the frequency of cesarean delivery compared with a low-dose regimen. In this review, high-dose regimens were associated with a higher rate of tachysystole, with similar maternal and neonatal complications in both regimens.

Rohn and colleagues studied the implementation of a low-dose protocol at a tertiary academic medical center.¹¹¹ This was a retrospective cohort study that evaluated cesarean delivery and chorioamnionitis before and after implementation of a low-dose oxytocin protocol. The protocol before the change directed that oxytocin should be initiated at 2 mU/min and increased by 3 mU/min every 15 minutes. This protocol was changed such that oxytocin was initiated at 2 mU/min and increased by 2 mU/min every 30 minutes. Evaluation of this change demonstrated no difference in cesarean delivery rates (14.8% vs. 15.2%, $P = .6$) but a significant increase in chorioamnionitis even after adjusting for confounders (1.27; 95% CI, 1.07 to 1.50; $P = .006$).

Limited data are available to guide how long the oxytocin infusion should be maintained once an adequate contraction pattern has been achieved. The authors of a meta-analysis¹¹² assessed the effect of discontinuation of oxytocin after onset of the active phase of labor. Eight studies were included and involved 1232 subjects. These researchers observed decreased cesarean delivery in subjects who had oxytocin discontinued (OR, 0.51; 95% CI, 0.35 to 0.74). Similarly, cases of nonreassuring fetal heart patterns were less likely among women who did not receive oxytocin after establishment of the active phase of labor (OR, 0.63; 95% CI 0.41 to 0.97).

Several experts have suggested that implementation of a standardized protocol is desirable to minimize errors in oxytocin administration.^{113–115} Clark and colleagues¹¹³ implemented an oxytocin checklist-based protocol at a tertiary-care facility and evaluated outcomes in 100 women before utilization of the protocol versus another 100 women after the protocol was put into practice. In the checklist-managed group, the maximum dose of oxytocin used to achieve delivery was significantly lower. No differences were noted before or after the protocol in the length of labor, total time of oxytocin administration, or rate of operative vaginal or cesarean delivery. When the protocol was then implemented throughout the Hospital Corporation of America system (125 obstetric facilities in 20 states), the rate of cesarean delivery decreased from 23.6% to 21.0%, and Apgar scores below 8 and newborn complications requiring NICU admission also decreased.

Overall, high-dose protocols have been associated with shorter labor and less chorioamnionitis, whereas low-dose regimens are associated with decreased tachysystole. The overall cesarean delivery rate has been shown to be similar between high- and low-dose protocols. At this time, no protocol has demonstrated its superiority in both efficacy and safety over another. However, experts recommend that each hospital's labor and delivery unit develop guidelines and use a checklist for the safe administration of oxytocin. Different hospital types with different staffing models may warrant different dosing regimens.

Combination Methods for Induction/Cervical Ripening

Multiple studies have evaluated the efficacy of using both mechanical and pharmacologic agents in combination.^{81,116–124} Studies comparing Foley/oxytocin concurrently with Foley alone have had mixed results, with two studies demonstrating a shorter time to delivery with Foley/oxytocin combined^{117,119} and other studies finding no difference.^{116,118} Studies comparing misoprostol/Foley concurrently versus misoprostol alone have also yielded mixed results,^{120,123} although a meta-analysis of six trials demonstrated a shorter time to delivery with the combination of misoprostol/Foley concurrently. One recent trial included four different methods of cervical ripening, including two combination methods, thereby allowing a direct comparison of combination methods.⁸¹ This was a four-armed randomized trial that evaluated the efficacy of misoprostol alone, Foley alone, misoprostol/Foley concurrently, and Foley/oxytocin concurrently. The investigators found a 4-hour shorter time to delivery for the combination methods compared with single-agent methods regardless of parity. Importantly, however, the shorter time to delivery for the combination of Foley/oxytocin was no longer statistically significant in models that censored for cesarean delivery and adjusted for parity. Therefore the investigators concluded that concurrent misoprostol/Foley was the superior method, achieving a delivery that was almost twice as fast as misoprostol alone or Foley alone (HR, 1.92; 95% CI, 1.42 to 2.59 and HR 1.87; 95% CI, 1.39 to 2.52), respectively. Although differences in the frequency of cesarean delivery were not statistically significant, multiparous women had a cesarean rate that was four times higher with Foley/oxytocin compared with misoprostol/Foley in this study (16% vs. 4%, $P = .27$). The point estimate for the frequency of maternal morbidity was also higher in the Foley/oxytocin group compared with misoprostol/Foley alone, as follows: chorioamnionitis: 16% versus 12.2%, $P = .22$; composite maternal morbidity (endometritis, third- or fourth-degree laceration, blood transfusion, wound separation/infection, readmission): 8% versus 4.1%, $P = .26$. Furthermore, use of an oxytocin infusion typically requires higher nursing ratios on most labor and delivery units compared with PG use. A meta-analysis from 2019¹²⁵ that included 11 trials ($n = 1869$) compared the combination of Foley and misoprostol with misoprostol alone and supported that the use of combined Foley/misoprostol resulted in no difference in cesarean risk (RR 0.95, 95% CI 0.80 to 1.13) but confirmed a 4-hour shorter time to delivery (-4.53 hours, interquartile range -5.79 to -3.27 hours), a lower risk of NICU admission (RR 0.71, 95% CI 0.53 to 0.96), lower risk of meconium (RR 0.62, 95% CI 0.43 to 0.90), a lower tachysystole with fetal heart rate changes (RR 0.53, 95% CI 0.30 to 0.94), and similar risk of endometritis (RR 1.07, 95% CI 0.43 to 2.61).

These data support misoprostol/Foley as the combination method of choice for cervical ripening from both a safety and efficacy standpoint. Data regarding efficacy and safety of single agent versus combination agent methods for induction in the setting of a prior cesarean delivery are not available.

ALTERNATIVE INDUCTION METHODS

Amniotomy

Artificial rupture of membranes may be used as a labor induction method, particularly when a cervix is favorable. Data are lacking to allow any recommendations about amniotomy alone as a method for labor induction, especially in the setting of an unfavorable cervix. However, there are data regarding amniotomy timing during labor induction when other agents such as oxytocin are already being used. A randomized trial by Macones and colleagues¹²⁶ found that early amniotomy (defined as <4 cm) in nulliparous women undergoing induction resulted in a shortened time to delivery, decreasing it by approximately 2 hours. It also increased the proportion of women who delivered within 24 hours (68% vs. 56%). There was no increase noted in complications with early amniotomy.¹²⁶ A recent systematic review and meta-analysis of seven trials ($n = 1775$),¹²⁷ including the one from Macones and colleagues,¹²⁶ demonstrated that early amniotomy had a shorter time to delivery (mean difference -3.62 hours, 95% CI -0.09 to -1.16) overall and a 5-hour shorter time to delivery when limiting the analysis to nulliparous patients (mean difference -5.12 hours, 95% CI -8.47 to -1.76). There was no difference in the rate of cesarean delivery or infectious morbidity.

Nipple Stimulation

Nipple stimulation has been evaluated in several studies in low-risk women. A systematic review of six trials comparing nipple stimulation with no intervention among women with unfavorable cervixes showed that stimulation significantly decreased the frequency of women who were not in labor within 72 hours (62.7% vs. 93.6%; RR, 67; 95% CI, 0.60 to 0.74).¹²⁸ No adverse outcomes were noted. Although the data are limited, nipple stimulation can be considered in an effort to initiate labor in low-risk women.

FAILED INDUCTION OF LABOR

Vaginal delivery is the goal of the induction process. Although no universal standard exists for what constitutes a “failed” induction, investigators have propounded that a key principle is to allow adequate time for cervical ripening and development of the active phase of labor before diagnosing a failed induction. The optimal time in the latent phase should be a balance between the chance that a vaginal delivery will ultimately be achieved and the chance that excess morbidity will accrue. *Failed induction* is a term that should be reserved for patients being induced who never progress to active labor.

Numerous studies have evaluated the association between the duration of the latent phase of labor and obstetric outcomes to determine the conditions under which a failed induction should be diagnosed.^{129–131} There is ample evidence that at least 12 hours of oxytocin administration after membrane rupture is essential before calling an induction failed.^{132–134} However, questions arise as to whether 12 hours is sufficient and the maximum length of time after oxytocin administration and membrane rupture that should elapse before diagnosing a failed induction.

A study from the Maternal-Fetal Medicine Units (MFMU) network by Grobman and colleagues from 2018¹³⁵ sought to define failed induction for nulliparous women using data from a large obstetric cohort from 25 hospitals. Among the 10,677 women who were evaluated, 96.4% had reached active phase of labor by 15 hours after cervical ripening had been completed, oxytocin had been initiated, and amniotomy had occurred. This finding was regardless of the indication for induction and/or whether premature rupture of membranes

(PROM) had occurred. Among the 3.6% of women who had not entered the active phase by 15 hours, 42% had a vaginal delivery. However, several maternal morbidities, including postpartum hemorrhage and chorioamnionitis, increased as the length of the latent phase increased. Importantly, there was no specific time point at which the chance of morbidities suddenly increased, but rather they increased continuously as a function of length of latent phase.

The initial impetus for these guidelines and studies that have evaluated the definition of failed induction of labor was to provide some guidance for a time threshold at which recommending a cesarean was reasonable given increased maternal and neonatal morbidity and decreased likelihood of vaginal birth once that time had been reached.

A recent review by Ayala and Rouse¹³⁶ reported on various studies and definitions of failed induction of labor and noted that although there are no randomized trials that compared failed labor induction criteria, the observational data consistently demonstrate the importance of at least 12 to 18 hours of oxytocin after membrane rupture before considering an induction of labor to have failed due to an individual not progressing beyond the latent phase (i.e., as long as maternal and fetal status allows).

This prior body of evidence should be distinguished from the recommendations from the 2014 NICHD, SMFM, and ACOG workshop on the prevention of primary cesarean delivery.¹³⁷ They proposed that *failed induction* be defined as failure to generate regular contractions approximately every 3 minutes and failure of the cervix to change after at least 24 hours of oxytocin administration. This timing recommendation of 24 hours was based on oxytocin duration alone and does not take into account the timing of membrane rupture or the time of cervical ripening in the diagnosis of a failed induction.

Taking all the evidence into account, membrane rupture and oxytocin administration should be a prerequisite before diagnosis of failed induction of labor. No absolute time cutoff for defining failed induction seems possible, given there will always be some trade-off between additional vaginal deliveries achieved and additional morbidities incurred with continued labor induction. On the basis of current data, it is reasonable to wait at least 15 to 18 hours in latent labor after oxytocin and the artificial rupture of membranes (AROM) before diagnosing a failed induction. Extending the time beyond 18 hours should be individualized and must take into account other factors to determine the risk/benefit balance in continuing labor.

OTHER

Premature Rupture of Membranes

In the setting of PROM at term, defined as rupture of membranes before the onset of labor, labor induction is often recommended because, as found by a meta-analysis of 23 randomized trials, it has been shown to decrease the time from PROM to delivery and the risk of chorioamnionitis, endometritis, and NICU admission without increasing cesarean delivery.¹³⁸ In the landmark and largest trial by Hannah and colleagues,¹³⁹ 5041 women with PROM at term were randomly assigned to receive intravenous oxytocin, vaginal PGE₂ gel, or expectant management for up to 4 days. Those randomized to the expectant management group were induced if complications such as chorioamnionitis developed. The rates of neonatal infection and cesarean delivery were not statistically different between the groups. Rates of clinical chorioamnionitis were lower in the group that received IV oxytocin. When Cochrane reviewers¹⁰⁶ compared oxytocin alone with vaginal PGs in 14 trials for labor induction after

PROM, both medications were found to be efficacious and can be considered in this clinical setting. However, some studies have shown an increase in chorioamnionitis with PG use.¹³⁹

Infection has been a concern with mechanical methods of ripening (i.e., Foley balloon catheter) in the setting of term PROM, but there are insufficient data to make specific recommendations against this practice. Two recent trials evaluated the utility of a Foley balloon catheter in addition to oxytocin for PROM. One multicenter trial¹⁴⁰ found no difference in time to delivery but a significantly increased risk of chorioamnionitis with the addition of the balloon catheter (8% vs. 0%, $P < .01$). Another trial among nulliparous women¹⁴¹ showed no difference in time to delivery (13 hours vs. 10.8 hours, $P = .9$) or risk of chorioamnionitis (10% vs. 5%, $P = .31$) with the addition of the Foley catheter, although this study may have been underpowered to detect clinically relevant differences.

Thus the current available data for term PROM suggest a benefit with induction of labor, generally with oxytocin, if labor has not begun by time of presentation. At this time, without compelling evidence that the addition of a Foley catheter is beneficial and with the suggestion that it may increase risk of chorioamnionitis, mechanical ripening with an intracervical balloon catheter is not generally recommended for induction in women with PROM.

Midtrimester Induction

In circumstances such as an in utero fetal demise or termination of pregnancy, options for delivery include induction of labor and dilation and evacuation (D&E). Although in the case of an intrauterine demise, some women may prefer expectant management in order to avoid an induction of labor, this approach raises concerns regarding the development of a consumptive coagulopathy and/or intrauterine infection. Although some studies¹⁴² report that 80% to 90% of women will spontaneously labor within 2 weeks of a fetal demise, the latency period may be longer. Although both induction of labor and D&E are considered acceptable, the latter has been associated with fewer complications.^{143–147} In one retrospective study in which women undergoing surgical termination between 13 and 23^{6/7} weeks of gestation were compared with women undergoing labor induction, those undergoing D&E were found to have an overall lower rate of complications than that in those undergoing induction (15% vs. 28%, $P = .02$).¹⁴⁴ The most commonly noted complication among women undergoing labor induction was retained placenta, which occurred in up to 21% of women. However, this conclusion may not be generalizable to circumstances when induction occurs with mifepristone combined with misoprostol. Randomized trial data^{148,149} suggest that a regimen of mifepristone plus misoprostol is superior in effecting a faster time to delivery and a lower risk of retained placenta compared with misoprostol alone. Because the accessibility of mifepristone may be logistically challenging, misoprostol is preferred if mifepristone is not available. Box 14.4 demonstrates various acceptable regimes for medical abortion. Finally, medical abortion is the preferred mode for delivery if a trained D&E provider is not available.

When an induction of labor is being planned, gestational age plays a significant role regarding the methods of induction. When the gestational age is less than 28 weeks, the uterus is less sensitive to oxytocin; therefore PGs or mechanical devices may be required to commence labor. Current induction protocols vary by dose, route, and gestational age (see Table 12.3). It is important to keep in mind that although side effects (uterine tachysystole, nausea, vomiting, diarrhea) and safety remain important considerations for the patient in these circumstances, fetal well-being is no longer an issue.

BOX 14.4 Regimens for Second-Trimester Medical Abortions

- Mifepristone, 200 mg, administered orally, followed in 24 to 48 hours by:
 - Misoprostol, 800 µg, administered vaginally, followed by 400 µg administered vaginally or sublingually every 3 hours up to a maximum of five doses.^a
 - Misoprostol, 400 µg, administered buccally every 3 hours up to a maximum of five doses also may be used.
- If mifepristone is not available:
 - Misoprostol, 400 µg, administered vaginally or sublingually every 3 hours up to five doses.^a Vaginal dosage is superior to sublingual dosage for nulliparous women.
 - A vaginal loading dose of 600 to 800 µg of misoprostol followed by 400 µg administered vaginally or sublingually every 3 hours may be more effective.
- If misoprostol is not available:
 - Oxytocin 20 to 100 units infused intravenously over 3 hours, followed by 1 hour without oxytocin to allow diuresis. Oxytocin dosage may be slowly increased to a maximum of 300 units over 3 hours.^b

^aIf the abortion is not complete after five doses, the woman may be allowed to rest for 12 hours before starting the cycle again.

^bHigh-dose oxytocin is not commonly used in the second trimester because of the inefficient response of the uterus to oxytocin during this gestational period.

From American College of Obstetricians and Gynecologists. Second-trimester abortion. Practice Bulletin No. 135. *Obstet Gynecol.* 2013;121:1394-1406.

Women with a prior cesarean delivery are candidates for induction of labor with PG in these circumstances as well. A 2009 review by Berghella and colleagues¹⁵⁰ reported an incidence of uterine rupture of 0.4%, hysterectomy 0%, and transfusion 0.2% for women undergoing second-trimester misoprostol terminations.

Outpatient Induction

Multiple large studies performed in the inpatient setting have demonstrated the safety and efficacy of the Foley catheter as a cervical ripening method.^{66,80–84} On the basis of the safety profile, the Foley catheter seems an ideal candidate for cervical ripening in the outpatient setting.^{80,86,87,151} Small studies have demonstrated feasibility and even suggested a decreased risk of cesarean when cervical ripening is done in this fashion.^{152,153} Numerous studies have also demonstrated improved patient satisfaction and birth experience for outpatient versus inpatient cervical ripening.^{87,154–157} In fact, when they were evaluating patient preferences for inpatient versus outpatient cervical ripening with methods including the Foley catheter, Howard and colleagues¹⁵⁷ found that women were willing to accept an extra 1.4 trips to the hospital and over an hour in additional travel time in order to be able to have outpatient as opposed to inpatient cervical ripening. Evidence also suggests fewer inpatient hours^{84,87,153,158} and decreased cost^{159–161} when cervical ripening occurs as an outpatient. Although no large studies have been conducted comparing the efficacy and safety of outpatient versus inpatient cervical ripening, it is reasonable to consider this option in low-risk women. This approach is supported by ACOG, which states that “outpatient use may be appropriate in carefully selected patients. Mechanical methods may be particularly appropriate in the outpatient setting.”¹

KEY POINTS

- Induction of labor should be undertaken when the benefits of delivery to either mother or fetus outweigh the risks of pregnancy continuation. Induction of labor for medical indications should not be delayed based on cervical examination status.
- The American College of Obstetricians and Gynecologists Hypertension Task Force has recommended delivery at or beyond 37^{0/7} weeks for mild gestational hypertension or preeclampsia without severe features rather than expectant management to improve maternal outcomes.
- Non—medically indicated induction of labor refers to the initiation of labor in an individual with a term pregnancy who is free of medical or obstetric indications; this should not occur before 39 weeks' gestation. It is reasonable to discuss induction of labor at 39 weeks as risk reducing in the applicable patient populations.
- Labor induction between 41^{0/7} weeks and 42^{0/7} weeks should be considered for women who remain undelivered by this gestational age. Induction after 42^{0/7} weeks and no later than 42^{6/7} weeks is recommended to reduce the risk of perinatal morbidity and mortality.
- The combination of shorter maternal height, increasing body mass index, nulliparity, gestational age older than 40 weeks, and modified Bishop score were the five predictors of cesarean delivery among women induced with intact membranes and an unfavorable cervix. A calculator to aid in patient counseling can be found online at <http://www.uphs.upenn.edu/labor-induction-calculator/> and, when implementing into care once a patient has presented for induction, has demonstrated a decrease in maternal morbidity.
- Many types of cervical ripening agents are effective. If a combination method is chosen, data suggest that misoprostol/Foley should be the combination method of choice given the shorter time to delivery after adjustment, suggestion of a lower risk of morbidity, and fewer nursing staff needs with misoprostol over oxytocin.
- Membrane rupture and oxytocin administration should be a prerequisite before diagnosis of failed induction of labor. On the basis of current data, it is reasonable to wait at least 15 to 18 hours in latent labor after oxytocin and rupture of membranes before diagnosing a failed induction. Care must be individualized and must take into account other factors to determine the risk/benefit balance in continuing labor.
- In patients with term PROM, Foley catheter placement is generally not recommended and the current available data suggest a benefit with induction of labor (vs. expectant management), generally with oxytocin, if labor has not initiated by time of presentation.
- No oxytocin protocol has demonstrated its superiority in both efficacy and safety over another. Each hospital's labor and delivery unit should develop guidelines and use a checklist for the safe administration of oxytocin. Different hospital types with different staffing models may warrant different dosing regimens.
- Options of delivery for midtrimester stillbirth or induction for pregnancy termination include induction of labor and D&E. The decision for which mode of delivery to choose must be individualized by practitioner experience, gestational age, and a patient's desires.

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